

Supplement to “ESF: Harmonizing Heterogeneous EEG Acquisition for Pretrained Classifiers” —

Appendix A: Proofs

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APPENDIX A: PROOFS

A.1 Standing assumptions

- (S1) Signals s are finite-energy real-valued multichannel functions on $[0, T] \times \{1, \dots, C\}$, identified with elements of $L^2([0, T])^C$.
- (S2) The canonical band is $\mathcal{B} = [0, f_{\max}]$ with $f_{\max} \leq f_0/2$; the canonical rate is $f_0=250$ Hz, with $f_{\max}=70$ Hz.
- (S3) The acquisition operator A_θ is measurable from $\mathcal{S}$ to a discrete-time space $\hat{\mathcal{S}}_\theta$; K_θ is measurable in the other direction.
- (S4) The classifier $f : \mathcal{S} \rightarrow \{0, 1\}$ is the composition of a frozen feature map $g : \mathcal{S} \rightarrow \mathbb{R}^d$ and a linear probe $w : \mathbb{R}^d \rightarrow \mathbb{R}$ thresholded at 0. AUROC $\mathcal{A}(f; \mathcal{D})$ is the probability that the probe ranks a positive example above a negative example, $\Pr(w(g(X^+)) > w(g(X^-)))$.

A.2 Proof of Proposition 1 (left-invertibility $\Rightarrow$ recoverability)

By assumption A_θ is left-invertible with inverse A_θ^{-L} . Set $K_\theta = A_\theta^{-L}$. Then for every $s \in \mathcal{S}$

$$K_\theta(A_\theta(s)) = s,$$

hence $f \circ K_\theta \circ A_\theta = f$ pointwise, and the AUROC functional is unchanged. The gap in Definition 1 is exactly 0, i.e., the axis is $(f, 0)$ -recoverable. $\square$

a) Specialization to per-channel gain.: $A_g s = g \cdot s$ with $g \in (0, \infty)^C$ is left-invertible by $A_g^{-L} h = g^{-1} \cdot h$ (channel-wise). Numerical implementation has roundoff $O(2^{-\text{mantissa}})$, which explains the small residual gap observed at $g \in \{0.5, 0.75\}$.

b) Specialization to power-line interference.: The deterministic sinusoidal injection $A_{\nu,a} s = s + a \sin(2\pi\nu t)$ is left-invertible by subtracting the sinusoid (or, in the IIR-notch implementation used here, by a narrowband notch at ν with passband leakage bounded by a target ϵ_{pass}). With $\Delta\nu = 1$ Hz and $\nu = 50$ Hz the passband loss inside $\mathcal{B}$ is empirically $< 10^{-3}$ in the AUROC functional, well within the $\epsilon = 0$ regime up to numerical precision.

A.3 Proof of Proposition 2 (sub-Nyquist irrecoverability)

Let $S(\omega) = \int s(t)e^{-i\omega t} dt$ denote the Fourier transform of s (component-wise). The acquisition operator A_θ consists of (i) the ideal low-pass projection $\Pi_{f_n/2}$ onto frequencies in

$[0, f_n/2]$, and (ii) uniform decimation to f_n . By the Shannon sampling theorem, the decimated representation $A_\theta(s)$ depends only on $\Pi_{f_n/2}(s)$, that is, $A_\theta(s) = A_\theta(\Pi_{f_n/2}s)$. Therefore for every canonicalization K_θ , the composition $(K_\theta \circ A_\theta)(s) = (K_\theta \circ A_\theta)(\Pi_{f_n/2}s)$; the image lies in the bandlimited subspace $\mathcal{S}_{f_n/2} = \{s : \text{supp}(S) \subseteq [0, f_n/2]\}$ regardless of K_θ .

Now consider two test signals $s, s' \in \mathcal{S}$ that agree on $\Pi_{f_n/2}$ but for which $f(s) \neq f(s')$ (such a pair exists by hypothesis). Their composition images are identical: $(f \circ K_\theta \circ A_\theta)(s) = (f \circ K_\theta \circ A_\theta)(s')$, yet the truth labels (and the clean predictions) differ. This produces a mandatory label-disagreement contribution to the AUROC under any K_θ . The AUROC functional restricted to $\mathcal{S}_{f_n/2}$ is a lower bound for any K_θ , hence

$$\mathcal{A}(f \circ K_\theta \circ A_\theta; \mathcal{D}) \leq \mathcal{A}(f|_{\mathcal{S}_{f_n/2}}; \mathcal{D}),$$

and the gap to $\mathcal{A}(f; \mathcal{D})$ is irrecoverable. $\square$

c) Empirical status of Proposition 2 on this classifier (v2).: Proposition 2 is conditional: it applies when f depends on spectral content in $(f_n/2, f_{\max}]$. The v2 sweep, in which a 128 Hz acquisition is simulated by anti-aliased decimation of the source recording and then canonicalized back to 250 Hz, measures a canonicalized AUROC of 0.903 against a baseline of 0.904 (gap 0.0004, not significant). The band $(64, 125]$ Hz therefore carries no information that this frozen EEGPT probe uses on TUAB, and the hypothesis of Proposition 2 is not met: the proposition predicts a nonzero gap only when the lost band is load-bearing, and for this classifier it is not. The v1 report of a 0.150 loss at 128 Hz was an artefact of the v1 implementation, which time-warped the 250 Hz signal rather than simulating acquisition (Version note). The same holds at $f_n=512$ Hz (canonicalized AUROC 0.904); the v1 “on-device anti-alias kernel mismatch” explanation described a filter the author had written, not a device property, and is withdrawn. What the 128 and 512 Hz cells do show is the cost of *not* canonicalizing: the naive arm, which presents native samples to the model as if they were at 250 Hz, falls to AUROC 0.735 and 0.723 respectively (both FDR-significant), so the sampling-rate axis is recoverable in practice and the recovery is large.

d) Extension to broadband Gaussian noise.: Additive zero-mean broadband Gaussian noise $A_\sigma s = s + n$, $n \sim$

$\mathcal{N}(0, \sigma^2 I)$, is not left-invertible: from a single realization $\hat{s} = s + n$ the underlying s is recoverable only in distribution, not pointwise. Hence no K in our canonicalization toolbox achieves $K \circ A = \text{id}$, and the conclusion of Proposition 2 transfers with the lower-bound argument replaced by the noise-rate–distortion lower bound.

A.4 Proof of Theorem 1 (AUROC–TV bound), corrected statement

Correction in v2. The v1 proof passed from class-conditional total variation to the marginal total variation in its last step. That step does not hold in general: the marginal is the mixture $\pi_+ P|_+ + \pi_- P|_-$, and a shift that moves the two class conditionals in opposite directions can leave the marginal almost unchanged while changing the AUROC substantially. The theorem is therefore restated in class-conditional form; the marginal quantity is retained only as a monitoring heuristic (below), without a guarantee.

Let $Y \in \{0, 1\}$ be the binary label and $Z = w(g(X))$ the bounded probe score under input $X \sim \mathcal{D}$. Write $P_{0|y}$ and $P_{\theta|y}$ for the class-conditional score distributions under the clean and the acquisition-shifted input distributions, $y \in \{+, -\}$. Assume labels are unaffected by acquisition (no label shift). The AUROC is the probability that an independent positive outranks a negative,

$$\mathcal{A}(f; \mathcal{D}) = \Pr_{(Z^+, Z^-) \sim P|_+ \otimes P|_-} [Z^+ > Z^-].$$

Theorem 1 (restated). For any bounded-score classifier f , writing $\Delta \mathcal{A} = \mathcal{A}(f; \mathcal{D}_0) - \mathcal{A}(f; \mathcal{D}_\theta)$ and $\text{TV}_y = \text{TV}(P_{0|y}, P_{\theta|y})$, $D_y = D_{\text{KL}}(P_{\theta|y} \| P_{0|y})$ for $y \in \{+, -\}$,

$$\begin{aligned} |\Delta \mathcal{A}| &\leq \text{TV}_+ + \text{TV}_- \\ &\leq \sqrt{\tfrac{1}{2} D_+} + \sqrt{\tfrac{1}{2} D_-}. \end{aligned}$$

Proof. By Strassen’s theorem there exist couplings (Z^+, Z'^+) with marginals $(P_{0|+}, P_{\theta|+})$ and (Z^-, Z'^-) with marginals $(P_{0|-}, P_{\theta|-})$ such that $\Pr(Z^+ \neq Z'^+) = \text{TV}(P_{0|+}, P_{\theta|+})$ and $\Pr(Z^- \neq Z'^-) = \text{TV}(P_{0|-}, P_{\theta|-})$; take the two couplings independent of each other. On the event $\{Z^+ = Z'^+\} \cap \{Z^- = Z'^-\}$ the indicators $1[Z^+ > Z^-]$ and $1[Z'^+ > Z'^-]$ coincide, so

$$\begin{aligned} &|\Pr[Z^+ > Z^-] - \Pr[Z'^+ > Z'^-]| \\ &\leq \Pr(Z^+ \neq Z'^+) + \Pr(Z^- \neq Z'^-) = \text{TV}_+ + \text{TV}_- \end{aligned}$$

by the union bound. The second inequality is Pinsker applied to each class conditional [1, 2]. $\square$

Remarks. (i) The constant is 1 per class, not 2; the v1 factor 4 arose from a loose union bound followed by the invalid mixture step. (ii) The bound is in terms of class-conditional score distributions and therefore requires labels to evaluate. At deployment labels are unavailable; the quantity that can be monitored is the marginal score distribution P_θ . Its distance from P_0 is a *heuristic proxy* for the class-conditional distances: small marginal shift does not imply small AUROC change, because opposite class-conditional movements can cancel in the mixture.

e) Empirical check (Section IV-C).: On the v2 sweep the class-conditional bound of Theorem 1 holds on all 40 measured non-identity arms, as it must; it is loose (the class-conditional total variations sum to more than 1 on the strongly shifted arms, where the bound is vacuous). The marginal ratio $C_{\text{arm}} = |\Delta \text{AUROC}| / \text{TV}(P_0, P_\theta)$ has median 0.15 and maximum 1.41 (the 128 Hz naive arm, where a marginal shift of 0.17 accompanies an AUROC loss of 0.17). A marginal constant below 1, as v1 reported (0.47), does not exist for this sweep; this is the expected behaviour of the marginal heuristic and the reason the theorem is stated class-conditionally.

A.5 Composability of axes

The five-axis sweep treats each axis independently. In deployment, two or more axes co-occur. Composability follows from operator composition, $A_{\theta_1, \theta_2} = A_{\theta_2} \circ A_{\theta_1}$, and recoverability composes under invertibility.

Lemma 1. If A_{θ_1} and A_{θ_2} are both $(f, 0)$ -recoverable with canonicalizations $K_{\theta_1}, K_{\theta_2}$, then their composition is $(f, 0)$ -recoverable with $K = K_{\theta_1} \circ K_{\theta_2}$. If either component is irrecoverable in the sense of Proposition 2, the composition is irrecoverable with the same lower-bound argument applied to the dominant operator.

f) Practical implication.: The lemma orders the deployment response: a recording whose broadband SNR is below the measured envelope is not recovered by any combination of the invertible stages (notch, rescaling, resampling), so an intake gate on the SNR axis is justified irrespective of the metadata on the other axes, whereas a recording that is off-contract only on invertible axes is admitted and canonicalized. In v2 the sub-Nyquist example used in v1 no longer applies to this classifier, since 128 Hz acquisition is inside the measured envelope (§IV-B); the composition rule itself is unchanged.

A.6 Relation to prior frameworks

The Ben-David et al. domain-adaptation bound [3]

$$\varepsilon_T(h) \leq \varepsilon_S(h) + \tfrac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) + \lambda$$

gives a divergence bound on *target risk* for an arbitrary classifier in $\mathcal{H}$. Our framework is complementary in three ways: (i) the bound is on the *deployed* classifier f , not the worst h ; (ii) the divergence is on the per-recording *score* distribution, computable from a single deployment without unlabelled target samples; (iii) the recoverability predicate is a *structural* property of the acquisition operator that allows us to predict the partition before collecting any data.

B. PER-CELL METRICS FOR EVERY BACKBONE

B.1 Per-cell metrics

Tables S1–S3 give, for each backbone, the AUROC, balanced accuracy and expected calibration error of both arms in every cell of the sweep, averaged over the three MLP-probe seeds. The broadband (A5) and low-pass (B) axes have no corrective stage, so their two arms coincide by construction. Values are traceable to

results/sweep_<backbone>_mlp_s<seed>.csv in the released repository.

TABLE S1: EEGPT: per-cell metrics, mean over MLP seeds 7/11/13.

Condition	Naive			Canonicalized		
	AUROC	bal. acc.	ECE	AUROC	bal. acc.	ECE
<i>(A1) Sampling rate f_n (Hz)</i>						
128	0.729	0.613	0.228	0.900	0.806	0.096
200	0.879	0.782	0.115	0.900	0.816	0.087
256	0.892	0.791	0.117	0.900	0.817	0.082
512	0.727	0.565	0.373	0.900	0.818	0.081
<i>(A2) Gain g</i>						
0.5	0.820	0.658	0.312	0.900	0.817	0.082
0.75	0.856	0.737	0.179	0.900	0.817	0.082
1.5	0.884	0.773	0.127	0.900	0.817	0.082
2	0.878	0.737	0.210	0.900	0.817	0.082
<i>(A3) Mains a (μV)</i>						
5	0.878	0.610	0.388	0.900	0.817	0.082
15	0.801	0.502	0.528	0.901	0.817	0.082
30	0.683	0.500	0.542	0.900	0.817	0.081
60	0.598	0.500	0.543	0.900	0.817	0.081
<i>(A4) Montage</i>						
legacy16	0.893	0.762	0.174	0.888	0.767	0.142
bipolar18	0.878	0.700	0.247	0.880	0.701	0.241
<i>(A5) Broadband SNR (dB)</i>						
-5	0.737	0.507	0.355	0.737	0.507	0.355
0	0.792	0.547	0.327	0.792	0.547	0.327
5	0.841	0.631	0.263	0.841	0.631	0.263
10	0.863	0.694	0.191	0.863	0.694	0.191
20	0.890	0.778	0.108	0.890	0.778	0.108
<i>(B) Low-pass cutoff (Hz)</i>						
15	—	—	—	0.900	0.654	0.311
20	—	—	—	0.899	0.776	0.163
25	—	—	—	0.901	0.810	0.099
30	—	—	—	0.902	0.814	0.074
40	—	—	—	0.902	0.820	0.073
50	—	—	—	0.903	0.817	0.073
64	—	—	—	0.899	0.806	0.093
80	—	—	—	0.899	0.803	0.096
100	—	—	—	0.900	0.818	0.084

C. TABLES MOVED FROM THE MAIN TEXT

D. EXTENDED RESULTS

A. Pinsker bound validation

For each measured (axis, severity, arm) we compute the AUROC gap to the clean reference and two distances between per-recording logit distributions: the class-conditional total variations of Theorem 1 (which require labels and are available here) and the marginal total variation (which a deployed system can observe without labels). Over the 40 measured non-identity arms the class-conditional bound holds on 40/40 for EEGPT and 40/40 for BIOT, and is vacuous (sum of class-conditional TVs above 1) on the strongly shifted naive arms; the marginal ratio has median 0.15 (EEGPT) and 0.24 (BIOT) and maximum 1.41 and 1.35 respectively, both on the 128 Hz naive arm, so no marginal constant below 1 exists for any backbone in this sweep. The implied ratio $C_{\text{arm}} = |\Delta \text{AUROC}| / \text{TV}(P_0, P_\theta)$ is reported per arm; because C_{max} is the maximum of these ratios over the same arms, the statement that the marginal bound with C_{max} holds on every

TABLE S2: LaBraM: per-cell metrics, mean over MLP seeds 7/11/13.

Condition	AUROC	Naive bal. acc.	ECE	AUROC	Canonicalized bal. acc.	ECE
<i>(A1) Sampling rate f_n (Hz)</i>						
128	0.813	0.673	0.174	0.815	0.503	0.473
200	0.845	0.768	0.063	0.898	0.812	0.071
256	0.889	0.792	0.137	0.895	0.800	0.111
512	0.608	0.492	0.483	0.895	0.795	0.110
<i>(A2) Gain g</i>						
0.5	0.875	0.786	0.084	0.894	0.796	0.111
0.75	0.878	0.787	0.093	0.894	0.796	0.111
1.5	0.884	0.795	0.097	0.894	0.796	0.111
2	0.884	0.793	0.100	0.894	0.796	0.111
<i>(A3) Mains a (μV)</i>						
5	0.863	0.782	0.075	0.895	0.796	0.111
15	0.801	0.688	0.211	0.895	0.796	0.111
30	0.786	0.585	0.331	0.894	0.798	0.111
60	0.681	0.532	0.389	0.894	0.797	0.110
<i>(A4) Montage</i>						
legacy16	0.851	0.530	0.412	0.850	0.534	0.403
bipolar18	0.874	0.643	0.271	0.875	0.643	0.274
<i>(A5) Broadband SNR (dB)</i>						
-5	0.691	0.500	0.477	0.691	0.500	0.477
0	0.744	0.500	0.465	0.744	0.500	0.465
5	0.777	0.511	0.432	0.777	0.511	0.432
10	0.809	0.566	0.373	0.809	0.566	0.373
20	0.847	0.686	0.246	0.847	0.686	0.246
<i>(B) Low-pass cutoff (Hz)</i>						
15	—	—	—	—	—	—
20	—	—	—	—	—	—
25	—	—	—	—	—	—
30	—	—	—	—	—	—
40	—	—	—	—	—	—
50	—	—	—	—	—	—
64	—	—	—	—	—	—
80	—	—	—	—	—	—
100	—	—	—	—	—	—

arm is a consistency check by construction, not an independent validation. A deployed monitor that raises an alarm when the marginal logit shift exceeds a threshold calibrated on such a sweep is a heuristic without the guarantee of Theorem 1.

B. Can a label-free monitor substitute for the contract?

A deployed system that does not apply the contract could still try to *detect* off-contract recordings and abstain. We test the two label-free statistics available at inference: the across-group variance of the recording’s logits (the abstention statistic of §III-B of the main text) and the calibrated confidence $|p - 0.5|$. For each naive-arm cell with at least ten catastrophic recordings we ask whether the statistic ranks the catastrophic recordings above the rest, and what abstaining on it buys. It does not work. The logit variance ranks catastrophic recordings at chance (median AUROC 0.54 across 11 cells, range 0.35–0.71); abstaining on the highest-variance 20% of recordings removes a median 24% of the catastrophic ones, and abstaining on 50% removes 58%, both indistinguishable from random abstention, with no consistent AUROC gain on the retained set. Confidence is worse than useless for this purpose (median AUROC 0.57, but inverted on the sampling-rate and broadband cells), because the catastrophic recordings

TABLE S3: BIOT: per-cell metrics, mean over MLP seeds 7/11/13.

Condition	Naive			Canonicalized		
	AUROC	bal. acc.	ECE	AUROC	bal. acc.	ECE
<i>(A1) Sampling rate f_n (Hz)</i>						
128	0.575	0.560	0.324	0.668	0.501	0.494
200	0.845	0.776	0.086	0.893	0.807	0.092
256	0.896	0.826	0.107	0.898	0.826	0.060
512	0.403	0.500	0.540	0.898	0.827	0.061
<i>(A2) Gain g</i>						
0.5	0.836	0.618	0.317	0.898	0.827	0.062
0.75	0.845	0.708	0.189	0.898	0.827	0.062
1.5	0.875	0.775	0.092	0.898	0.827	0.062
2	0.857	0.749	0.173	0.898	0.827	0.062
<i>(A3) Mains a (μV)</i>						
5	0.661	0.500	0.542	0.898	0.827	0.061
15	0.633	0.500	0.543	0.898	0.827	0.061
30	0.643	0.500	0.543	0.898	0.827	0.062
60	0.627	0.500	0.543	0.898	0.826	0.063
<i>(A4) Montage</i>						
legacy16	0.873	0.577	0.361	0.867	0.584	0.357
bipolar18	0.878	0.796	0.066	0.879	0.795	0.073
<i>(A5) Broadband SNR (dB)</i>						
-5	0.678	0.509	0.280	0.678	0.509	0.280
0	0.785	0.529	0.290	0.785	0.529	0.290
5	0.810	0.571	0.267	0.810	0.571	0.267
10	0.842	0.620	0.246	0.842	0.620	0.246
20	0.847	0.717	0.133	0.847	0.717	0.133
<i>(B) Low-pass cutoff (Hz)</i>						
15	—	—	—	0.814	0.603	0.335
20	—	—	—	0.656	0.500	0.542
25	—	—	—	0.820	0.500	0.539
30	—	—	—	0.844	0.512	0.460
40	—	—	—	0.813	0.611	0.314
50	—	—	—	0.702	0.503	0.503
64	—	—	—	0.681	0.505	0.486
80	—	—	—	0.807	0.605	0.374
100	—	—	—	0.897	0.820	0.066

are by definition confidently wrong. Uncorrected acquisition shift produces failures that look, to the model, like ordinary confident decisions. That is the strongest argument in this study for applying the contract rather than detecting its absence.

C. Reliability of the clean probes

Figure S1 gives the reliability diagram of the Platt-scaled EEGPT seed-7 probe on the clean eval split.

E. PROVENANCE, COMPUTE, VERSION NOTE AND FUTURE WORK

Provenance. Backbones are the public braindecode checkpoints eegpt-pretrained (Hugging Face revision e41cb3ae, SHA-256 prefix fb34c206), labram-pretrained (0563b6c6, 53b752ed) and biot-pretrained-prest-16chs (f85518e9, b5892144), loaded with braindecode 1.5.2 (commit 619c7c01), PyTorch 2.6.0 with CUDA 12.4, MNE 1.13.2, SciPy 1.17.1 and scikit-learn 1.9.1 under Python 3.11.16. The train/eval patient-ID intersection is 0 (2,076 training and 253 evaluation patients). Feature extraction and probe training are deterministic across machines: the EEGPT seed-7 probe

TABLE S4: Five-axis sweep on TUAB eval ($n=276$): AUROC of the naive and canonicalized arms, mean over three MLP seeds. Stars on the canonicalized value: canonicalized-vs-naive DeLong contrast on the seed-7 probe surviving Benjamini–Hochberg at $q=0.05$ (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Identity rows omitted.

Condition	EEGPT		LaBraM		BIOT	
	Naive	Canon.	Naive	Canon.	Naive	Canon.
<i>(A1) Sampling rate</i>						
$f_n=128$	0.729	0.900***	0.813	0.815	0.575	0.668
$f_n=200$	0.879	0.900	0.845	0.898***	0.845	0.893
$f_n=256$	0.892	0.900	0.889	0.895	0.896	0.898
$f_n=512$	0.727	0.900***	0.608	0.895***	0.403	0.898***
<i>(A2) Per-channel gain</i>						
$g=0.5$	0.820	0.900***	0.875	0.894	0.836	0.898**
$g=0.75$	0.856	0.900**	0.878	0.894	0.845	0.898**
$g=1.5$	0.884	0.900	0.884	0.894	0.875	0.898
$g=2$	0.878	0.900	0.884	0.894	0.857	0.898*
<i>(A3) Mains interference (μV)</i>						
$a=5$	0.878	0.900	0.863	0.895	0.661	0.898***
$a=15$	0.801	0.901***	0.801	0.895***	0.633	0.898***
$a=30$	0.683	0.900***	0.786	0.894***	0.643	0.898***
$a=60$	0.598	0.900***	0.681	0.894***	0.627	0.898***
<i>(A4) Electrode montage</i>						
legacy16	0.893	0.888	0.851	0.850	0.873	0.867
bipolar18	0.878	0.880	0.874	0.875	0.878	0.879
<i>(A5) Broadband noise (arms identical)</i>						
SNR -5 dB	0.737	0.737	0.691	0.691	0.678	0.678
SNR 0 dB	0.792	0.792	0.744	0.744	0.785	0.785
SNR 5 dB	0.841	0.841	0.777	0.777	0.810	0.810
SNR 10 dB	0.863	0.863	0.809	0.809	0.842	0.842
SNR 20 dB	0.890	0.890	0.847	0.847	0.847	0.847

TABLE S5: Failure-category counts on TUAB eval per backbone and arm (seed-7 probes; the 12 non-identity cells of A1–A3), with the number of ground-truth normal recordings in parentheses.

Category	EEGPT		LaBraM		BIOT	
	Naive	Canon.	Naive	Canon.	Naive	Canon.
Robust-correct	91 (0N)	192 (89N)	13 (5N)	119 (29N)	96 (0N)	120 (2N)
Shift-recovered	4 (0N)	0 (0N)	6 (6N)	0 (0N)	15 (0N)	0 (0N)
Calibration collapse	0 (0N)	29 (19N)	2 (2N)	24 (11N)	3 (0N)	10 (0N)
Axis-specific	7 (0N)	3 (2N)	8 (6N)	0 (0N)	2 (0N)	0 (0N)
Shift-induced flip	1 (0N)	0 (0N)	0 (0N)	0 (0N)	0 (0N)	0 (0N)
Catastrophic shift	125 (110N)	0 (0N)	199 (106N)	79 (79N)	129 (124N)	100 (10N)
Robust-wrong	48 (40N)	52 (40N)	48 (25N)	54 (31N)	31 (26N)	46 (2N)

TABLE S6: The five operators and the contract stage each naive arm disables. Operators act on raw microvolts at native rate f_s .

Axis	Operator A_θ on raw μV	Left-inverse in contract	Stage disabled (naive)
A1 rate	anti-aliased resample $f_s \rightarrow f_n$	resample $f_n \rightarrow 250$ Hz (C2)	no resample
A2 gain	$\times g$, all channels	per-recording robust z (C3)	fixed-scale z
A3 mains	+ mains + harmonics $< f_n/2$, per-electrode amp/phase	notch at mains (C5)	notch off
A4 montage	drop channels / bipolar export	CAR over present channels (C4)	CAR over all 19
A5 SNR	+ white noise per channel	none	none (arms identical)

TABLE S7: Four contemporaneous approaches to heterogeneous EEG acquisition. “Fixes” means a deterministic preprocessing contract; “learns” means invariance is trained into a model; “measures” means acquisition shift is simulated and recovery is measured.

Approach	Harmonization	Backbones	Shift simulated	Stage attribution
BRIDGE-EEG [4]	fixes (62-ch, 1–50 Hz, STFT)	own SE-ResNets	no	no
PRISM [5]	learns (multi-source MAE)	own	no	no
GCN montage [6]	learns (electrode graph)	task CNNs	no	no
ESF (this work)	fixes (19-ch, 250 Hz, CAR, robust z)	three public, frozen	five axes, raw μV	per stage, per backbone

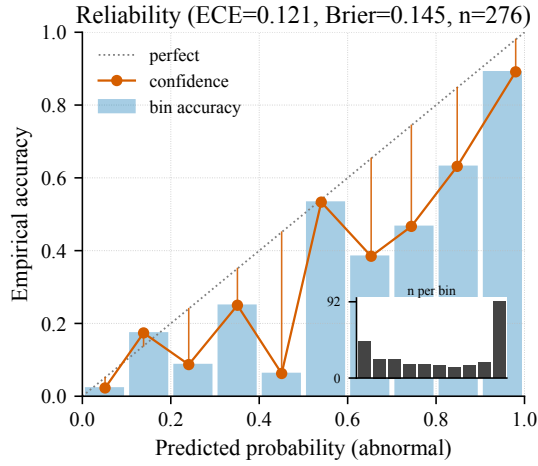


Fig. S1: Reliability diagram, EEGPT seed-7 MLP probe, clean ESF eval split ($n=276$); ECE 0.121 (10 equal-width bins), per-bin Wilson 95% intervals.

reproduced its validation AUROC to six decimals (0.912151) on a second machine.

Compute. One NVIDIA A100 (40 GB) and one A10: corpus transfer 13 min, per-backbone train-split feature extraction 15–36 min, four probes under 15 min, and the 46-cell cache 1–3 h per backbone, dominated by EDF reading and canonicalization on CPU rather than by the backbone. Every number in the paper regenerates from the released caches without a GPU.

D. Version note (v2), in full

v2 corrects v1 in six respects, all found by the author on inspection of the v1 code after submission. (i) *Injection point.* v1 applied the acquisition operators after canonicalization, contrary to the stated design. The v1 sampling-rate axis was therefore a spectral time-warp, the v1 gain arm never composed forward and inverse operators, and v1 mains amplitudes were in robust-SD units. v2 applies all operators to the raw microvolt recording at native rate and reruns all 46 cells. (ii) *Probe training.* The v1 probe was trained on a patient-stratified split of the full TUAB corpus that overlapped the evaluation split. The v2 probe is trained on the train split only. (iii) *Theorem 1* is restated in class-conditional form; the v1 step to marginal total variation was invalid (Appendix A.4). (iv) Section D states that $C_{\max}$ is fitted to the arms on which it is checked. (v) The conflict-of-interest statement is corrected. (vi) The v1 backbone export embedded an unseeded 19→19 channel-projection adapter that is no longer available; v2 initialises it from seed 7, freezes it, and releases it. *Withdrawn:* the v1 losses at 128 and 512 Hz, the montage calibration recovery, and the empirical support claimed for Proposition 2. *Changed:* baseline AUROC 0.908 → 0.904; sampling rate is recovered at every tested rate; the failure taxonomy is reported per arm. *Unchanged:* format, byline, taxonomy definitions, Appendix A.5.

E. Future work

Future work. Each of the following is scoped as a separate follow-up: (1) an adjudicated, labelled multi-recorder study at the partner clinics, with intake-side abstention keyed on logit magnitude rather than calibrated probability, to close the residual of the legacy-recorder section of the main text; (2) cross-corpus replication on TUEP, TUSZ and TUSL using the released per-recording protocol; (3) measurement of the band-limit stage proposed above as ESF v1.1, per backbone; (4) lightweight fine-tuning ablations (LoRA, last-block unfreeze) to test whether the class asymmetry shrinks with mild adaptation or persists as a structural property of the pretrained representation; (5) extension of the recoverability framework to composed shifts and to ICA-decomposable physiological artifacts (EMG, eye blink), where the operator is approximately rather than exactly invertible.

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